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MILITARY SPECIFICATION

PROPORTIONERS, MECHANICAL FOAM LIQUID

1. SCOPE

1.1 Scope.- This specification covers the requirements applicable to the design and construction of positive displacement rotary type proportioners driven by positive displacement rotary type water motors, for the purpose of introducing foam liquid into five mains at carefully controlled rates depending upon the flow of water through the equipment.

1.2 Classification.- Mechanical foam proportioners shall be of the following classes as specified (see 6.1):

Class FP-180
Class FP-1000

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-F-555 - Foam-Forming Liquids, Concentrated Fire Extinguishing, Mechanical.
FF-B-171 - Bearings, Ball, Annular (General Purpose).
QQ-N-281 - Nickel-Copper-Alloy (Monel and R-Monel) Bars, Plates, Rods, Sheets, Strips, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (K-Monel).
GGG-P-781 - Puller and Puller Kit, Mechanical, and Mechanical Puller Attachment.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed Waterproofed, Flexible.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-P-197 - Preservation, Packaging and Packing of Antifriction Bearings, Associated Parts, and Subassemblies.
MIL-S-890 - Steel: Forgings and Bars for Hulls, Engines, and Ordnance (Heat Treated).
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact), Shipboard Application, Tests for.
MIL-D-963 - Drawings, Production, Procedure for Procurement of.
MIL-D-5480 - Data, Engineering and Technical: Reproduction Requirements for.
MIL-P-15024 - Plates, Identification—Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).
MIL-P-15137 - Provisioning and Technical Documentation for Repair Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-P-15424 - Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.
MIL-B-16261 - Bronze, Bearing; Castings.
MIL-M-16576 - Metal, Gun: Castings.
MIL-N-17163 - Nickel-Copper Alloy, Wrought; (55-60 Percent Nickel) Low Permeability.
MIL-C-17516 - Copper-Silicon Alloy Bars, Plates, Rods, Sheets, Strips, Flat Wire, Wire, Forgings, and Structural and Special Shaped Sections.
MIL-F-20042 - Flanges, Pipe, Bronze (Silver Brazing).
MIL-D-70327 - Drawings, Engineering and Associated Lists.

STANDARDS

MILITARY

MIL-STD-7 Types and Definitions of Engineering Drawings.

FSC 4210

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MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.

DRAWINGS

BUREAU OF SHIPS
B-153 - Packing and Gaskets, Application of.
B-214 - Root Connections for Attaching Piping
810-1385891 - Foam Liquid Pick-up Tube.

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS
A-286-57T - Quenched and Tempered Alloy Steel Bars (Tentative).
B-143-52 - Tin-Bronze and Leaded Tin-Bronze Sand Castings.
B-144-52 - High-Leaded Tin-Bronze Sand Castings.
B-164-49T - Nickel-Copper Alloy Rods and Bars (Tentative).

AERONAUTICAL MATERIAL SPECIFICATION
AMS-4616 - Silicone Bronze.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 2, Pa.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Qualification.- The material furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list (see 6.2).

3.2 General requirements.

3.2.1 Design and construction.- The design and construction of the proportioners shall be the lightest and most compact possible consistent with the requirements of reliability, accessibility to repair, resistance to corrosion and erosion, and economy. The principle of reliability, however, is paramount and no compromise of this principle shall be made with the other basic requirements of design. The proportioners shall be of such design that they will operate over a long period of years with a minimum of wear or erosion of operating parts, such as castings, liners, end plates, rotors, gears, shafts, glands and bearings. The parts subject to unavoidable wear or erosion shall be of the best material available for the purpose intended so as to reduce these detrimental effects to a minimum.

3.2.1.1 The proportioners shall operate in a manner satisfactory to the bureau or agency concerned when inclined as follows:

(a) Up to 15 degrees to either side (permanently inclined).
(b) With the ship rolling up to 60 degrees from the vertical to either side.
(c) With the ship pitching 10 degrees up and down from the normal horizontal plane.

3.2.1.2 Each proportioner shall consist of a horizontal positive displacement rotary foam liquid pump driven by a positive displacement rotary water motor through a solid sleeve type coupling. The pump shall take suction from a foam liquid tank and shall discharge to the downstream side of the water motor.

3.2.1.3 A separately driven foam liquid booster pump shall not be required.

3.2.1.4 Each proportioner shall be designed so that the volumetric ratio of the foam liquid pumped, to the sum of the water flowing through the water motor plus the foam liquid pumped, shall be not greater than 7-1/2 percent (0.075) nor less than 5 percent (0.050).

3.2.2 Shockproofness.

3.2.2.1 The design of the proportioners shall be such that they are capable of passing the class HI (high-impact) shock test specified in Specification MIL-S-901.

3.2.2.2 Shockmounts are not to be used unless bureau or agency approval of the mount and its proposed application is given.

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3.2.2.3 The determination of stresses resulting from the shock tests is accomplished by the application of the usual methods of analytical mechanics. The allowable stress in components due to shock shall be the yield strength (0.2 percent offset) of the material. Normal stresses in parts, such as the tensile stress in a bolt due to initial pull-up, shall not be added to the shock stresses. It is of paramount importance that the maximum accelerations possible under the shock tests be utilized in the design of holding down bolts, supporting feet or lugs, and main structural members of the equipment, and that no failure or permanent deformation occur as a result of shock. Components of an assembly mounted on a common rigid base need not be designed for higher accelerations than the values for which the whole assembly is designed unless the designer has reason to believe that higher accelerations will be experienced by the components.

3.2.2.4 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance, as experience indicates that large clearances allow impacting and subsequent failure.

3.2.3 Alignment.-

3.2.3.1 Alignment between separate components of a unit shall be maintained by means of keyways or other adequate means.

3.2.3.2 The design of all proportioners shall be such that alignment will not be disturbed or undue stresses set up in any part by normal vibration, contraction and expansion of piping attached thereto.

3.2.3.3 In no case shall a piece of equipment be rigidly supported from more than one plane.

3.2.4 Noise and vibration.- The design and construction of the proportioners shall be such that they will operate with a minimum of noise and vibration at all speeds.

3.2.5 Threaded parts.- Threaded parts, such as bolts, nuts and studs shall conform to Handbook H28. In general, threads shall conform to the course-thread series, unless the use of the fine thread series should show a definite advantage. The use of tap bolts or cap screws is prohibited unless specifically approved by the bureau or agency concerned where the use of through bolts or studs is impracticable.

3.2.6 Packing and gaskets.-

3.2.6.1 All packing and gaskets shall be in accordance with Drawing B-153.

3.2.6.2 Mechanical seals, packing glands and packing shall be of a design and type suitable for the intended purpose and as approved by the Bureau or agency concerned. Mechanical seals are preferred.

3.2.7 Root connections.- Root connections for attaching pipes shall be in accordance with Drawing B-214.

3.2.8 Mountings.- Proportioners shall be designed for horizontal mounting.

3.2.9 Bedplates.- Bedplates, when used, shall be sufficiently rigid to permit handling, shipment, and installation of the units onboard ship, without disturbing the alignment of the assembled units; and such that the normal distortion, weaving or vibration of the supporting structures onboard ships cannot cause misalignment between the foam pumps and water motors.

3.2.9.1 All bearing and seating surfaces of bedplates shall be finish machined.

3.2.9.2 Each component part of an assembled unit supported directly by a bedplate shall be doweled thereto to facilitate reassembly and maintenance of alignment.

3.2.10 Bearings.-

3.2.10.1 Radial and thrust bearings shall be of the ball type.

3.2.10.2 The use of ball bearings shall be subject to the speed and load limitations specified in Specification FF-B-171.

3.2.10.3 Bearings shall be provided on both ends of all rotor shafts, and shall be sealed in a positive manner to prevent loss of the lubricant. In addition to the lubricant seal for each bearing, a second seal assembly shall be provided to separate the bearing from the liquid handled. The space between the two seals shall be provided with a drain to permit continuous drainage to atmosphere.

3.2.10.4 All thrust bearings installed shall be suitable for counteracting any unbalanced hydraulic or mechanical thrust in either direction. Consideration shall be given to the fact that rolling, pitching or listing of a ship at sea may introduce thrust loads even though the unit is in hydraulic balance.

3.2.10.5 Ball bearings shall be lubricated in accordance with Specification FF-B-171.

3.2.10.6 Ball bearing races shall be locked to prevent rotation.

3.2.11 Casings and heads.-

3.2.11.1 Casings shall be made as light as possible, consistent with strength, and shall be ribbed

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and braced on the outside as necessary to give stiffness and rigidity under service conditions. The interiors shall be smooth cored with easy bends wherever the direction of flow is changed.

3.2.11.2 Fireman suction and discharge openings shall be cast integral with the casings and shall have flanged faces machined to a fine tool finish. All flanged connections shall be in accordance with Specification MIL-F- 20042.

3.2.11.3 The heads shall have accurately machined faces to a fine tool finish. Joints between casings and heads may be made up either metal to metal or with gaskets. Heads shall be either rabbeted and doweled or doweled to the casings.

3.2.11.3.1 Adequate provisions shall be made in the design of heads so that bearings may be readily inspected or renewed.

3.2.11.4 Forcing bolts shall be provided for breaking joints.

3.2.11.5 All proportioners shall be fitted with drain and vent connections.

3.2.11.6 Sufficient clearance shall be provided around bolt heads and nuts to permit the use of ordinary tools.

3.2.12 Rotors and shafts.-

3.2.12.1 All rotors shall be accurately finished machined or ground throughout. They shall be secured to the shaft by keys or other method suitable for the purpose intended. The design of the rotors shall be such as to reduce noise and vibration to a minimum under all load conditions.

3.2.12.1.1 Provision shall be made for rotating the rotors of the unit manually.

3.2.12.2 Means shall be provided for the escape of the liquid from all points where dangerously high pressures might develop due to trapping during operation, such as between gear teeth.

3.2.12.3 All shafts shall be accurately machined and ground.

3.2.12.4 Means shall be provided on the shaft of the water motor or the foam liquid pump so that a portable tachometer may be quickly and easily applied.

3.2.12.5 All shaft threading shall be counter to the direction of rotation.

3.2.12.6 The design shall permit positive location of each rotating assembly to prevent axial movement.

3.2.12.6.1 Each rotating assembly shall be dynamically balanced.

3.2.13 Timing gears.-

3.2.13.1 Timing gears shall be of the spur or single helical type, and shall be secured to the shafts by keys, or other method suitable for the purpose intended.

3.2.13.2 Timing gears shall be hardened to a minimum hardness of 42 Rockwell "C".

3.2.13.3 Timing gears shall be fully enclosed and adequately lubricated.

3.2.14 Identification plates.-

3.2.14.1 Identification plates shall be furnished on each proportioner in a readily accessible location and shall be type A, E or H(a) in accordance with Specification MIL-P-15024, except that identification plates of plastic or aluminum are not acceptable.

3.2.14.2 Identification plates shall be secured to equipment with corrosion resistant metallic screws.

3.2.14.3 Identification plates shall contain the following information:

(a) Manufacturer's name.
(b) Manufacturer's model or type and size.
(c) Service application.
(d) Manufacturer's serial number.
(e) Salient design characteristics:
  (1) Capacity (Total G.P.M.): Min.___, Max.___.
  (2) Pressures (P.S.I.G.): Normal___ (Min.___, Max.___) Hydrostatic test
  (3) Water Motor R.P.M.: Min.___, Max.___.
  (4) Foam solution strength: 5 to 7-1/2 percent.
(f) Contract number.
(g) Federal stock number.
(h) Section for inspector's stamp.

3.2.15 Welding.- Welding shall be in accordance with Standard MIL-STD-278.

3.2.16 Painting.- Proportioners are not to be painted.

3.3 Detail requirements.-

3.3.1 The proportioner shall maintain the volumetric ratio specified in 3.2.1.4 for all volumes of flow through the water motor from the minimum to the maximum ratings specified herein.

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3.3.1.1 Capacity ratings shall be the total discharge of water plus foam liquid measured in gallons per minute (g.p.m.) downstream of the proportioner.

3.3.1.2 Pressure ratings shall be fireman pressures measured in pounds per square inch (p.s.i.) at the water motor inlet.

3.3.1.3 Pressure drop shall be measured in p.s.i. across the water motor.

3.3.1.4 The foam liquid to be pumped shall be in accordance with Specification O-F-555.

3.3.2 Class FP-180 proportioners.-

3.3.2.1 Ratings.- Class FP-180 proportioners shall operate as specified at a capacity rating from 60 g.p.m. to 180 g.p.m. and at a pressure rating from 100 p.s.i. to 175 p.s.i. At a pressure of 150 p.s.i., the pressure drop shall not exceed 23 p.s.i. at 60 g.p.m. capacity, nor 33 p.s.i. at 180 g.p.m. capacity.

3.3.2.2 Connections.-

3.3.2.2.1 Inlet and outlet connections of the water motor shall be 2-1/2 inch hose thread conforming to fire-hose coupling threads NH in accordance with Handbook H28. The inlet connection shall be female and outlet connection shall be male.

3.3.2.2.2 Inlet and outlet connections shall be provided with protective caps. The caps shall be anchored to the proportioner body with chains and shall be of corrosion resisting material.

3.3.2.3 Foam liquid pumps.-

3.3.2.3.1 The suction of the foam liquid pump shall be fitted with a 3-way valve fitting, one branch of which is connected to the up-stream side of the water motor and the other two branches designed for the attachment of standard Navy type pick-up tubes. The valve shall be easily operable but shall have positive stops in each position so that vibration will not cause the valve to move from a set position. On one position the valve shall provide for taking water from the up-stream side of the motor through the pump for priming or flushing the pump; while the other two positions provide for taking foam liquid from either of the two foam liquid pick-up tubes.

3.3.2.3.2 A connection shall be provided between the discharge of the pump and the down-stream side of the water motor.

3.3.2.3.3 Each proportioner shall be furnished with two foam liquid pick-up tube assemblies as shown on Drawing 810-1385891.

3.3.2.4 Installation.- The proportioner shall be suitable for use either as a portable unit or for stationary installation using flexible inlet and outlet connections, and hence shall be provided with a base suitable for the purpose intended. The base and carrying frame shall be of nonsparking materials to prevent fire hazard if the unit is pulled along the deck.

3.3.2.4.1 The carrying frame shall provide means for holding the two pick-up tube assemblies, when they are not in use.

3.3.2.5 Dimensions and weight.- The overall dimensions of the proportioner, less carrying frame and pick-up tubes, shall not exceed 20 inches by 13 inches by 10 inches. The weight of the proportioner complete with inlet and outlet connection caps, pick-up tubes, and carrying frame shall not exceed 125 pounds.

3.3.3 Class FP-1000 proportioners.-

3.3.3.1 Ratings.- Class FP-1000 proportioners shall operate as specified at a capacity rating from 200 g.p.m. to 1000 g.p.m. and pressure rating from 100 p.s.i. to 175 p.s.i. At a pressure of 150 p.s.i., the pressure drop shall not exceed 20 p.s.i. at 200 g.p.m. capacity, nor 35 p.s.i. at 1000 g.p.m. capacity.

3.3.3.2 Connections.- Inlet and outlet connections of the water motor shall be 5 inch flanges cast integral with the casing.

3.3.3.3 Flushing.- To the upstream side of the water motor shall be attached at 3/4 inch gate valve and a flanged flexible hose coupling for flushing the foam liquid pump after it has pumped foam liquid.

3.3.3.4 Foam liquid pump.-

3.3.3.4.1 The foam liquid pump shall be self-priming and shall operate without cavitation. The foam liquid pump shall be designed for a maximum suction lift of 15 inches of mercury when handling 60 g.p.m. of the specified foam liquid.

3.3.3.4.2 The foam liquid pump suction shall be provided with a 2-inch flanged flexible hose coupling; while the discharge side shall be fitted with a three way spring loaded valve. In the normal position, the valve shall discharge foam liquid via a pipe connection to the downstream side of the water motor; while in the other position and only when manually held against a spring tension, the valve shall provide for recirculating the foam liquid back to the storage tank via an 1-1/4-inch flanged flexible hose coupling. Upon release of the handle, the spring shall return the valve to its normal position.

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3.3.3.4.3 A check valve shall be provided in the line between the foam liquid pump discharge and the water motor outlet.

3.3.3.5 Flanges.- All flanges shall conform to Specification MIL-F-20042.

3.3.3.5.1 The overall length of each flanged flexible hose coupling shall be 12 inches.

3.3.3.6 Installation.-

3.3.3.6.1 The proportioner shall be installed in a horizontal section of a 5-inch fire main approximately 100 feet from the fire pump. A quick opening valve will be installed in the fire main adjacent to and on the upstream side of the proportioner. The proportioner shall be designed sufficiently rugged so that installation in the fire main without flexible pipe connections will be satisfactory. Further, all proportioner components shall satisfactorily withstand full fire main pressure and any vibrations or shock loads incident to operation of the quick opening valve.

3.3.3.6.2 The unit shall be designed so that all bearings, rotors, and shafts may be removed and replaced without disconnecting the motor from the fire main.

3.3.3.7 Dimensions and weight.- The overall dimensions of the complete proportioner unit shall not exceed 30 inches by 20 inches by 16 inches, exclusive of the flanged flexible hose couplings. The weight of the complete proportioner, dry, including the flanged flexible hose couplings shall not exceed 500 pounds.

3.4 Material.- Material shall be as specified in table I. Material not definitely specified shall be of the best quality suited for the purpose intended, and shall be subject to review and approval by the bureau or agency concerned. No materials subject to corrosion from sea water or foam liquid shall be used in any part of the proportioners except ball bearings and timing gears. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product. Aluminum is not acceptable.

3.4.1 All bolts, nuts and studs shall be of corrosion resistant material.

3.5 Repair parts and tools.-

3.5.1 Onboard.- Applicable onboard repair parts shall be furnished and processed in accordance with Specification MIL-R-15137.

3.5.1.1 Each set of onboard repair parts shall consist of the parts specified in table II.

3.5.1.2 The pump and motor rotor assemblies shall be completely assembled ready for installation in the unit for which intended and shall include any nonrotating parts which can be installed only by removing some rotating part or parts of the pump or motor rotor.

Table I - Material.

Application	Material	Military[1,2] specification	Commercial[1,2] specification
Casings and heads	Gun metal	MIL-M-16576	ASTM B143-52 alloy 1B
Shafts	Monel, wrought K monel, wrought, or monel	MIL-N-17163 QQ-N-286 QQ-N-281	Inco-monel 403 K monel ASTM B164-49T Class A
Rotors	Gun metal	MIL-M-16576	ASTM B143-52, alloy 1B
Rotor housing	Gun metal	MIL-M-16576	ASTM B143-52, alloy 1B
Timing gears	Alloy steel	MIL-S-890	ASTM A286-57T class BC (AISI 4140)
Casing side wear plates	Bronze, bearing	MIL-B-16261	ASTM B-144-52 alloy 3A

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Table I - Material (cont'd.)

Application	Material	Military[1,2] specification	Commercial[1,2] specification
Packing glands	Gun metal	MIL-M-16576	ASTM B143-52, alloy 1B
Bolts, nuts, studs	Silicon bronze	MIL-C-17516	AMS-4616

[1]Materials used may be in accordance with either Military specifications or any recognized equivalent industry specifications, at the proportioner manufacturer's option. Commercial specifications referenced in the last column are representative but not exclusive, and others may be used, such as the following, when specifically approved by the bureau or agency concerned:

ASTM - American Society for Testing Materials.
SAE - Society of Automative Engineers.
AISI - American Iron and Steel Institute.

[2]Detail drawing and lists of material shall reference the material specifications actually followed in each case, and shall include the class, type or grade of material used in each case, as applicable.

Table II - Onboard repair parts.

Name of part	Quantity per set				
	1 - 4 Identical units per ship	5 - 8 Identical units per ship	9 - 12 Identical units per ship	13 - 16 Identical units per ship	17 - 20 Identical units per ship
Rotors, complete, including shafts, keys, nuts and couplings, complete sets	1	1	1	1	2
Ball bearings, complete sets	1	2	3	4	5
Timing gears, sets	1	2	2	3	3
Mechanical seals, complete sets	1	2	3	4	5
Gaskets and "O"-Rings, complete sets	2	4	8	12	16
Special packing, complete sets	2	4	8	12	16

3.5.1.3 The quantities of individual repair parts shall not be reduced because of duplication of parts furnished to make the pump or motor rotor assemblies.

3.5.1.4 Gaskets which may be fabricated from bulk material with hand tools shall not be furnished as repair items. The gaskets should be identified on the provisioning list so that the applicable bulk stock material number is assigned.

3.5.2 Tools.-

3.5.2.1 The equipment design shall require no special tools, if possible. However, where the design and construction is such as to require the use of special tools for proper maintenance of the proportioners, a complete set of special tools shall be included in each set of onboard repair parts. Special tools are defined as those tools not listed in the Navy Stock List of General Stores. (Copies of this catalog may be consulted in the Office of the Government inspector.).

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3.5.2.2 Tools which are available in the Navy *Stock List of General Stores*, such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of puller sets in accordance with Specification GGG-P-781 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools shall be detailed, and included in the list of material.

3.5.2.3 Each box containing onboard repair parts shall contain a list entitled "List of onboard repair parts and tools". The list shall be in a format suitable for use on drawings and in the technical manuals. It shall not be less than nominal 8-1/2 by 11 inches in size.

3.5.2.3.1 The format of the list shall include a heading and columns of data for the items listed. The heading shall include titles and applicable entries as follows:

Reproduced from drawing number_____
Number of ships_____
Application _____
Contract number_____
Manufacturer _____
Quantities are for_____units per ship.
The columns shall include:
Piece number_____
Name of part _____
Quantity_____
Drawing number _____
Federal stock number _____
Additional columns may be used as applicable.

3.5.2.4 The list shall be so treated as to be resistant to oil, water and fading.

3.5.3 Stock.- The selection, stock numbering and quantities of repair parts for stock shall be determined and processed in accordance with Specification MIL-R-15137.

3.5.3.1 Unless otherwise specified in the contract or order, the cognizant supply demand control point for repair parts for stock is Ships Parts Control Center, Mechanicsburg, Pennsylvania.

3.6 Drawings.-

3.6.1 Drawings shall be class 2 in accordance with Specification MIL-D-963 and MIL-D-70327. Drawings shall be engineering drawings of the following types as defined in Standard MIL-STD-7:

(a) Assembly drawing covering complete proportioner units including pump, driver, and accessories. This drawing shall include complete performance data and performance curves, and also a table of weights of components and complete unit.
(b) Detail assembly drawing of proportioner which shall include a sectional assembly with complete list of material, with references to applicable detail drawings. All running clearances shall be shown. The bill of material shall include an indication of each part to be furnished as an onboard repair part.
(c) Detail drawings of all parts necessary for the evaluation of the equipment, and of all parts necessary for installation, operation, maintenance, and overhaul of the equipment by Naval activities without assistance from the original supplier. Multi-detail drawings are preferred, but mono-detail drawings may be used.
(d) Envelope drawing. The requirements of this drawing may be combined with the assembly drawing.

3.6.1.1 The performance curves on the assembly drawing as originally submitted for approval may be calculated curves and shall be so indicated. The complete performance curves on the validated drawing shall be based on the performance acceptance tests and shall be so indicated. These curves shall show the relationship between pressure drop, motor and pump r.p.m., and percent fluid concentration for the pressure range from 100 p.s.i. to 175 p.s.i. at capacities of 60 g.p.m. and 120 g.p.m. for the FP-180 proportioner and at capacities of 200 g.p.m.; 500 g.p.m., 750 g.p.m. and 1000 g.p.m. for the FP-1000 proportioner.

3.6.1.2 The quantity of onboard repair parts shall be indicated on the detail assembly drawing for sets based on installations consisting of 1 to 4, 5 to 8, 9 to 12, 13 to 16 and 17 to 20 identical equipments per ship.

3.6.2 The certification data shall be furnished in accordance with Specification MIL-D-9630 listed and shall include the characteristics required in 3.2.14.3 (e) herein plus a list of onboard repair parts furnished under the contract or reference to such list appearing on applicable drawings.

3.6.3 Drawings and certification data sheets submitted for approval shall be blueprints or equal, type II, class 1 in accordance with Specification MIL-D-5480. They shall be submitted to the bureau or agency concerned for approval in accordance with Specification MIL-D-963.

3.6.4 Final drawings and certification data sheets (tracings) shall be in accordance with Specification MIL-D-963.

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3.7 Technical manuals.-

3.7.1 Technical manuals shall be type III in accordance with Specification MIL-M-15071.

3.7.2 At least two photographs of the completely assembled proportioner taken 180 degrees apart, to show as much detail as practicable, shall be included in the manual.

3.7.3 The technical manuals shall contain instructions for cleaning the proportioners of preservative compound when the equipment is installed.

3.7.3.1 The instructions in the technical manual shall be substantially as follows: When a proportioner is first put in operation the oil sump and gear case shall be filled with lubricating oil and the proportioner operated by hand for 10 minutes, after which the oil shall be completely drained out and discarded. The proper quantity of fresh oil shall then be put in. After each proportioner has been operated for 15 hours in normal usage, the oil shall be again drained and discarded, and refilled.

3.7.4 Unless otherwise specified (see 6.1) distribution of technical manuals shall be in accordance with Specification MIL-M-15071.

3.8 The requirements specified herein cover minimum performance characteristics and maximum size and weights. This specification is not intended to be restrictive as to design and materials, provided it is proved by actual tests that other designs and materials will result in equaling or bettering the requirements of this specification. If an alternate design is proposed, then all departures from specification requirements should be thoroughly described giving reasons and advantages therefor. Drawings submitted with the original requests need not be in accordance with the drawing specification requirements of this specification. However, such drawings are required prior to qualification approval.

3.8.1 Every deviation from design or material specified herein shall have specific approval of the bureau or agency concerned.

3.9 Workmanship.- The equipment, including all parts and accessories, shall be manufactured and finished with first class workmanship in all respects.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Tests requirements specified herein shall be at the expense of the contractor.

4.3 Classification of tests.- Tests shall be classified as follows:

(a) Qualification tests
(b) Shop tests
(c) Performance acceptance tests.

4.4 Qualification tests.- Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. The pressure drop, capacity and other characteristics of units for qualification tests shall be as arranged between the Bureau of Ships and the exhibitor. Qualification is required for each class.

4.4.1 Qualification tests shall be conducted as specified in 4.4.2, 4.4.3, 4.4.4 and 4.4.5 to determine that performance and design of the proportioners are in compliance with this specification. Minor deviations in design and materials may be allowed at the discretion of the Bureau of Ships.

4.4.2 The qualification tests shall adequately demonstrate the ability of the proportioner to handle the specified capacities between the specified limits of water motor inlet pressures. Sufficient data shall be taken during the tests to prepare characteristic curves showing the relationship between motor inlet pressure and the following:

(a) Pressure drop through the water motor.
(b) Motor r.p.m.
(c) Pump r.p.m.
(d) Percent foam liquid concentration downstream of the proportioner.

The above data shall be taken at pressure ratings of 100 p.s.i.g., 125 p.s.i.g., 150 p.s.i.g. and 175 p.s.i.g. at capacity ratings of 60 g.p.m., 120 g.p.m. and 180 g.p.m. for the class FP-180 proportioners, and at capacity ratings of 200 g.p.m., 500 g.p.m., 750 g.p.m. and 1000 g.p.m. for class FP-1000 proportioners.

4.4.3 Endurance test.- The proportioner shall be subjected to an endurance test of not less than 200 hours duration, with the unit operating at a pressure of 150 p.s.i.g. and maximum rated capacity while pumping full capacity of foam liquid. All

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data required by 4.4.2 for the specified conditions shall be recorded at the end of each 50 hours of operation. At the end of each 100 hours of operation, the operating pressure shall be reduced to 100 p.s.i.g. and the flow reduced to the minimum rated capacity for a period long enough to record data specified in 4.4.2. At the end of the 200 hours of operation and recording of all data required above, the working pressure shall be increased to 175 p.s.i.g. and the flow increased to the maximum rated capacity for a period of one hour, at the end of which, data specified by 4.4.2 shall be recorded. Any change in noise, vibration or leakage shall be noted during the 175 p.s.i.g. operation. The unit shall complete the endurance test without excessive noise vibration or wear of parts. Failure of any part of the proportioner during the endurance test shall be considered as failure of the proportioner to qualify. Failure of the proportioner to maintain the volumetric ratios specified in 3.3.1 shall be considered as failure of the proportioner to qualify.

4.4.4 Shock tests.-

4.4.4.1 The proportioner shall be subject to the high impact shock test of Specification MIL-S-901 as specified in 3.2.2.

4.4.4.2 Shock tests will be conducted at the manufacturer's plant. Where a manufacturer is unable to conduct these tests at his own plant, he may arrange to have them conducted at a commercial laboratory, or Government laboratory suitably equipped to conduct tests. Where shock tests are conducted at a Government laboratory, copies of the applicable class 2 drawings in accordance with Specification MIL-D-963 shall accompany the units.

4.4.4.3 Compliance with shock resistance requirements may be covered by test of a similar unit subject to bureau or agency concerned approval.

4.4.4.4 Equipment previously accepted will not be resubmitted for shock test except when evidence of low shock resistance develops in the units installed. The bureau or agency concerned will initiate action with the manufacturer for correction of deficiencies, and may require shock tests at any time.

4.4.4.5 The unit shall be considered to have failed to pass the shock tests in the event of any of the following:

(a) Breakage of any parts, including mounting bolts.
(b) Appreciable distortion or dislocation of any part, such as shaft, mounting feet, bearings.
(c) A mechanical unbalance of more than two times the amplitude of unbalance measured prior to tests at rated speed.

4.4.4.5.1 Units which have been subjected to the high-impact shock test and have failed to conform to the requirements of this specification will not be acceptable.

4.4.4.5.2 Units which have been subjected to the high-impact shock test and have successfully passed this test shall be considered acceptable for such service as the bureau or agency concerned may authorize. Such equipment shall be reconditioned by the manufacturer as follows:

(a) Minor deformations of mounting flanges shall be corrected.
(b) Minor deformations affecting alignment shall be corrected.
(c) All bearings shall be replaced.
(d) Each part shall be carefully examined by the contractor and any part which he considers substandard shall be replaced.

Minor deformations shall be defined as those which do not cause unqualified rejection of the design under high-impact shock test but which are in excess of the design dimensional tolerances specified on the applicable drawings.

4.4.5 Upon completion of the qualification tests, the unit shall be dismantled and thoroughly inspected for wear, malfunctioning or failure of parts, and all findings reported to the Bureau of Ships.

4.5 Shop tests.-

4.5.1 All motor and pump casings shall be tested hydrostatically at 300 p.s.i.

4.5.2 No castings shall be impregnated prior to the hydrostatic tests and no castings shall be impregnated after the hydrostatic tests without specific approval of the bureau or agency concerned.

4.5.2.1 If the contractor desires to seal porosity in castings by means of impregnation, the Government inspector shall require that the areas where leakage occurs be radiographed to determine the type of defect which is responsible for the leakage, and also to determine the internal soundness of the metal. The radiography, together with complete information as to the condition of the castings, including marked drawings and the percentage of castings requiring impregnation, shall be forwarded to the bureau or agency concerned with the request for permission to impregnate. Radiographs shall be produced in accordance with Standard MIL-STD-271.

4.5.2.2 Radiographic examination is required only on defective areas. All radiographic examination and impregnation shall be accomplished by the contractor.

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4.5.2.3 The methods, procedures and materials used in impregnation of castings will be specified by the bureau or agency concerned at the time of acting on requests to impregnate.

4.5.3 Each proportioner shall be run not less than 15 minutes at a pressure of 150 p.s.i.g. Motor speed, pump speed, and foam liquid concentration downstream of the water motor shall be recorded for each of the following capacity ratings: 60 and 180 g.p.m. for the class FP-180 proportioner and 200 and 1000 g.p.m. for the class FP-1000 proportioner. During this test, each unit shall meet the requirements specified herein.

4.5.3.1 The pressure drop through the motor at maximum rated capacity and 175 p.s.i.g. shall be noted and recorded.

4.5.4 All unusual noises shall be noted, investigated and corrected.

4.5.5 Any leakage shall be noted and corrected.

4.6 Performance acceptance tests.- (To be conducted at manufacturer's plant)

4.6.1 Prior to delivery of any equipment under a contract or order, sample proportioners shall be selected in accordance with table III for tests under operating conditions, including foam liquid, to develop characteristic curves as specified in 4.4.2.

Table III - Sampling for performance acceptance tests.

Number of proportioners being procured	Number of proportioners to be tested
1 - 8	1
9 - 15	2
16 - 40	3
41 - 110	4
111 - 300	5
301 - 400	6
401 - 600	7

4.6.1.1 The tests specified in 4.6.1 shall demonstrate that the pressure drop through the water motor, and that the volumetric ration of foam liquid pumped to the total discharge are within the limits required herein, and failure of the unit tested to meet the requirements shall be cause for rejection of equipments built to the same design as the unit tested. The required data may be determined by any convenient methods that will insure obtaining results with probable errors of not greater than 2 percent. All gages shall be calibrated before and after the tests, or oftener if necessary.

4.7 Reports.- Following the tests specified in 4.5 and 4.6, a comprehensive report, including curves specified in 4.6.1, shall be forwarded triplicate to the bureau or agency concerned.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of shipboard repair parts.-

5.1.1 Basic equipment.-

5.1.1.1 Preservation and packaging.- Preservation and packaging of the basic equipment shall be sufficient to afford adequate protection against corrosion deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by carrier and which will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping container or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's name and destination.

5.1.2 Onboard repair parts and tools.- Onboard repair parts and tools shall be preserved and packaged level A, packed level C and marked levels A and C in accordance with Specification MIL-R-196, MIL-P-197 and MIL-P-15424 as applicable. Where bin and drawer type storage is not provided (see 6.1), onboard repair parts shall be packaged level B. Marking shall then be level B for packing. Items not specifically covered therein shall be preserved and packaged in accordance with Specification MIL-P-116. Repair parts shall be individually packaged, except where used in sets.

5.2 Domestic shipment and storage or overseas shipment requirements.-

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1):

5.2.1.1 Cleaning, preservation and packaging.- The equipment shall be cleaned, preserved, and packaged in accordance with the guide lines of Specification MIL-P-116 and as specified herein.

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5.2.1.1.1 The proportioners need not be disassembled for preservation, but the following procedures shall be followed, after the shop tests:

(a) Thoroughly drain all oil from the gear case and oil sump.
(b) Fill oil sump and gear case with type P-2 preservative compound.
(c) Operate the proportioner long enough to thoroughly coat all oil lubricated parts and surfaces.
(d) Thoroughly drain off all preservative compound. This compound shall be disposed of and not reused.
(e) Refill oil sump and gear case with fresh P-2 preservative compound.
(f) Operate the proportioner long enough to thoroughly coat all oil lubricated parts and surfaces.
(g) Thoroughly drain off all preservative compound.
(h) Anti-friction bearings shall be filled to operating level with the normal operating lubricant.

5.2.1.1.2 All openings shall be covered with barrier material conforming to grade A of Specification MIL-E-121 and secured with pressure sensitive waterproof tape. Flange openings shall be covered with grade A barrier material and covered with blank flanges of mild steel, tempered hard board or waterproof plywood. Flanges shall be secured with not less than 4 bolts with nuts and lockwashers. Openings up to 2-1/2 inches in diameter, plastic plugs may be used in lieu of barrier materials and blank flanges.

5.2.1.2 Repair parts and tools.- Repair parts and tools shall be preserved and packaged level A in accordance with Specification MIL-R-196, MIL-P-197 and MIL-P-15424 as applicable. Items not specifically covered therein shall be preserved and packaged in accordance with Specification MIL-P-116. Unless otherwise specified (see 6.1) repair parts shall be individually packaged, except when used in sets.

5.2.1.3 Packing.-

5.2.1.3.1 Level A.- Proportioners, packaged as specified (see 6.1) shall be packed as follows: class FP-180 proportioners shall be individually packed in nailed wood or wood-cleated plywood boxes. Class FP-1000 proportioners shall be individually packed in sheathed crates conforming to Specification MIL-C-104. Wood and wood-cleated plywood boxes shall conform to Specification PPP-B-621 (class 2) and PPP-B-601 (overseas type) respectively. Wood or wood-cleated boxes shall be provided with sealed caseliners conforming to Specification MIL-L-10547 and the appendix thereto. Shipping containers shall be closed and strapped in accordance with the applicable container specification appendix thereto.

5.2.1.3.2 Level B.- Proportioners packaged as specified (see 6.1) shall be packed as follows: class FP-180 proportioners shall be individually packed in nailed wood or wood-cleated plywood boxes. Class FP-1000 proportioners shall be individually packed in sheathed crates conforming to Specification MIL-C-104. Wood and wood-cleated plywood boxes shall conform to Specification PPP-B-621 (class 1) and PPP-B-601 (domestic type) respectively. Shipping containers shall be closed in accordance with the applicable container specification or appendix thereto.

5.2.1.3.3 Repair parts (stock) and tools.- Repair parts and tools shall be packed level A or B as specified (see 6.1) in accordance with the requirements of the applicable specification referenced in 5.2.1.2.

5.2.1.4 Anchoring blocking, bracing and cushioning shall be in accordance with the appendix to Specification MIL-C-104.

5.2.1.5 Marking.- In addition to any special marking required by the contract or order interior packages and exterior shipping containers shall be marked in accordance with the requirements of the applicable packaging and packing specification and Standard MIL-STD-129.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class of foam liquid proportioner required (see 1.2).
(c) The quantity and distribution of technical manuals if different from that specified (see 3.7).
(d) Whether bin and drawer type storage is provided (see 5.1.2).
(e) Whether packaging, packing or marking requirements other than those specified in 5.1 are required (see 5.2).

6.1.1 Bid data.- The following information should be submitted with each bid:

(a) Manufacturer's model and type designation of foam proportioners.
(b) Sectional assembly of foam proportioner or reference to previous approved drawings.
(c) Drawing showing overall dimensions of the unit or reference to previous approved drawing.

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(d) A definitive priced list of onboard repair parts and special tools.
(e) Characteristic curves showing relationship between pressure drop, fluid concentration, pump revolutions per minute (r.p.m.), motor r.p.m., and inlet pressure over the working range.
(f) Weight of the complete unit.
(g) Manufacturers of major components such as water motors and foam liquid pumps.

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the times set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL 15208, whether or not such products have actually been so listed by that date. The attention of the supplier is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

6.3 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pa.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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